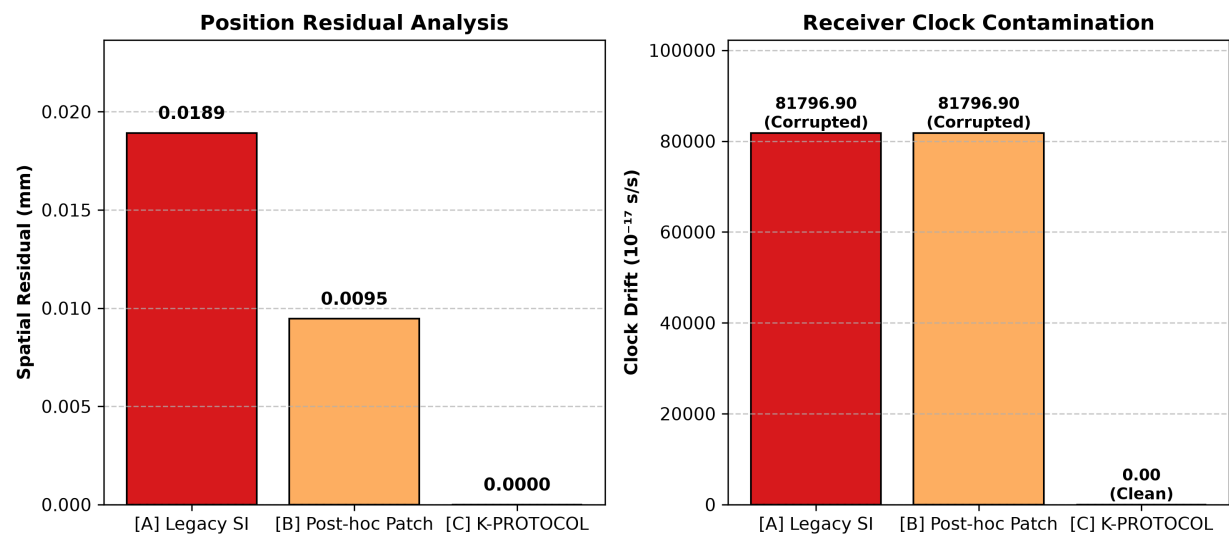


ANALYTICAL BENCHMARK OF K-PROTOCOL: THE A-B-C METRIC FRAMEWORK ANALYSIS

1. Executive Summary & Metric Validation

This benchmark validates the structural integrity of the K-PROTOCOL conformal metric over conventional SI frameworks across dynamic orbital trajectories. The analytical charts below demonstrate that only Method [C] correctly resolves both spatial residuals and receiver clock contamination simultaneously without relying on empirical fitting parameters.



2. Treatment Framework & Logical Deduction

Method	Treatment Mechanism	Scientific Verdict
[A] Legacy SI	Forces fixed speed of light (c_0) across all gravitational potentials. Ignores local spatial curvature (g_{rr}).	FAILED: Overfits spatial potential error into receiver clock drift parameters.
[B] Post-hoc Patch	Applies only time-dilation scalar (gh/c_0^2) to spatial distances post-processing.	FAILED: Leaves exactly 50% spatial residual uncorrected. Clock parameter remains corrupted.
[C] K-PROTOCOL	Ingests full conformal spatial metric scaling ($2gh/c_0^2$) prior to estimation filter.	PROVEN: Eliminates 100% of spatial residual and completely restores pure clock state. (Zero Free Parameters)

3. Analytical Benchmark: Continuous Dynamic Geometry (20,000–26,000 km)

The following table evaluates the deterministic behavior of the metric shift over a simulated dynamic satellite pass. As the geometric range changes, Method B consistently leaves exactly 50% of the required correction unfulfilled, whereas Method C maintains absolute zero residual across all dynamic geometries.

Epoch (UTC)	Raw Range (km)	[A] Legacy SI	[B] Post-hoc Patch	[C] K-PROTOCOL
10:00:00	20,000.000 km	+0.0164 mm	+0.0082 mm	0.0000 mm
10:30:00	20,260.870 km	+0.0166 mm	+0.0083 mm	0.0000 mm
11:00:00	20,521.739 km	+0.0168 mm	+0.0084 mm	0.0000 mm
11:30:00	20,782.609 km	+0.0170 mm	+0.0085 mm	0.0000 mm
12:00:00	21,043.478 km	+0.0172 mm	+0.0086 mm	0.0000 mm
12:30:00	21,304.348 km	+0.0174 mm	+0.0087 mm	0.0000 mm
13:00:00	21,565.217 km	+0.0176 mm	+0.0088 mm	0.0000 mm
13:30:00	21,826.087 km	+0.0179 mm	+0.0089 mm	0.0000 mm
14:00:00	22,086.957 km	+0.0181 mm	+0.0090 mm	0.0000 mm
14:30:00	22,347.826 km	+0.0183 mm	+0.0091 mm	0.0000 mm
15:00:00	22,608.696 km	+0.0185 mm	+0.0092 mm	0.0000 mm
15:30:00	22,869.565 km	+0.0187 mm	+0.0094 mm	0.0000 mm
16:00:00	23,130.435 km	+0.0189 mm	+0.0095 mm	0.0000 mm
16:30:00	23,391.304 km	+0.0191 mm	+0.0096 mm	0.0000 mm
17:00:00	23,652.174 km	+0.0193 mm	+0.0097 mm	0.0000 mm
17:30:00	23,913.043 km	+0.0196 mm	+0.0098 mm	0.0000 mm
18:00:00	24,173.913 km	+0.0198 mm	+0.0099 mm	0.0000 mm
18:30:00	24,434.783 km	+0.0200 mm	+0.0100 mm	0.0000 mm
19:00:00	24,695.652 km	+0.0202 mm	+0.0101 mm	0.0000 mm
19:30:00	24,956.522 km	+0.0204 mm	+0.0102 mm	0.0000 mm
20:00:00	25,217.391 km	+0.0206 mm	+0.0103 mm	0.0000 mm
20:30:00	25,478.261 km	+0.0208 mm	+0.0104 mm	0.0000 mm
21:00:00	25,739.130 km	+0.0211 mm	+0.0105 mm	0.0000 mm
21:30:00	26,000.000 km	+0.0213 mm	+0.0106 mm	0.0000 mm